

Security and Privacy Evaluation of Popular Augmented and Virtual Reality Technologies

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Abstract—Augmented (AR) and Virtual Reality (VR) have emerged across several industries and are steadily progressing to a comprehensive level of user adoption. However, while different companies are introducing AR/VR devices and applications, little or no attention is being placed on evaluating their security. To this aid, we collected the most commonly available 10 AR/VR devices and applications, including Hololens, Oculus, Google Glass, Valve Index, HTC Vive, Raptor AR, Psious (Amelia), Magic Leap, Epson Moverio, and IKEA Place AR. We then performed a security and privacy evaluation of the devices and their associated platforms while focusing on - device authentication, user profiling, access control, database security, and others. Our findings inform the limitations of these identified devices and applications and suggest areas where enhancements can be provided to ensure users' better, privacy-preserving, more secure experience.

Index Terms—AR, VR, Privacy, Cybersecurity.

I. INTRODUCTION

Augmented (AR) and Virtual Reality (VR) offer ways to enrich our user experience, giving us a sense of immersion [11], [22]. AR is “a real-time direct or indirect view of a physical, real-world environment that has been enhanced or augmented by adding virtual computer-generated information to it” [7]. VR is defined as “a medium composed of interactive computer simulations that sense the participant’s position and actions and replace or augment the feedback to one or more senses, giving the feeling of being mentally immersed or present in the simulation” [26]. These technologies are becoming more prevalent across several industries, including education, gaming, and healthcare. However, despite tremendous growth in various sectors, several aspects, such as security and privacy postures, are understudied.

Thus, to obtain a substantial understanding of AR/VR in the security and privacy realm, we analyzed 10 devices and applications from prominent industrial sectors, including *Hololens*, *Oculus*, *HTC Vive*, *Google glass*, *Valve Index*, *Raptor AR*, *Psious (Amelia)*, *Magic Leap*, *Epson Moverio*, and *IKEA Place AR*. We focused our analysis on webpage security, authentication, user profiling, access control, and database security. We examined the infrastructure of their supported devices and applications using online tools such as BuiltWith¹. Our analysis showed several security and privacy vulnerabilities. For instance, most of these technologies (n=7) do not provide two-factor authentication, and in some cases (n=4), there is no means for users’ account deletion. Finally, we summarize by providing recommendations to ensure a more secure and enhanced user experience.

¹<https://builtwith.com/>

II. RELATED WORK

AR/VR devices help users to attain a sense of immersion in a non-physical world or a mixture of both the physical and non-physical world [10], [22]. These devices render visual and auditory stimuli and provide image sensors that help track users’ movement, scene reconstruction, and other computer vision applications [3], [18]. Continuous developments in these devices suggest that they are at the beginning of their journey as the next generation computing platform [20].

Authentication in AR/VR devices has been introduced in different models. From Biometric authentication, which utilizes the distinctiveness of a user’s characteristics such as facial recognition, fingerprints, iris recognition, vein recognition, retina scanning, and voice recognition, to Knowledge-based authentication, such as pin and password that utilizes information possessed by the user, and Token-based authentication that leverages on the possession of an object such as smartcards by the user [10]. These authentication approaches have been strengthened with several schemes such as Two-Factor Authentication (2FA) and Multi-Factor Authentication (MFA). Luo et al., in their work, presented OcuLock, which leveraged biometric authentication with the use of eye gaze and recorded EER of 3.55% when tested against impersonation attacks and 4.97% on statistical attacks [21].

Langfinger et al. [15] in their work highlighted the importance of protecting the video streams in AR. They believed the video frames, when accessed by an attacker, could lead to the loss of valuable information such as architectural plant layout, passwords, secret formulas, and recordings of the device camera. Additionally, hackers could manipulate the video stream and display any form of indecent content to the user. Therefore, they proposed a cloud architecture to protect sensitive data from unauthorized access, which relies heavily on the edge server. Ahn et al. also worked on securing the visual output of AR systems through deep reinforcement learning using a fog computing node that continuously learns policies that enables the filtering of malicious content that could obstruct real-world objects during an AR experience [2].

Among other works, Lebeck et al. [16]. Introduced an OS framework, Arya, which focuses on the visual output of AR applications. In this framework, policies are introduced based on conditional situations; for instance, an action is performed when an AR object violates specific policies. An example, in AR-induced automotive windshields, virtual objects that obstruct pedestrians should be made transparent [2].

Additionally, a methodology was introduced that could be used to generate policies in securing visual output in AR by performing reinforcement learning and fog computing. As an extension of security in AR/VR devices, access control provides policies that determine access levels to data and information. Lehman et al. developed a framework called PrivacyManager targeted at mobile AR applications [19]. The framework allows developers to incorporate standard functionality on several existing AR applications. In addition, it ensures flexible access controls, such as issuing a warning when a pedestrian is nearing a crosswalk. These previous works have focused on enhancing existing functionalities and the development of new ones; however, there is less work focused on evaluating the currently available devices and applications. Our research would bridge this gap by analyzing significant security aspects of these devices.

III. METHOD

We conducted a full-scale security evaluation of top AR/VR devices and applications used in gaming, healthcare, engineering, education, accessibility, retail, energy, and entertainment. To begin with, the first and second authors individually performed an online search using Google Search Engine to collect the commonly available AR/VR devices and applications. The search includes keywords like “Augmented reality devices in gaming”, “Virtual reality devices in gaming”, “Augmented reality devices in education”, “Virtual reality devices in education”, and so on. Next, we combined the individual search results. We selected the most prominent devices and applications: Oculus Quest2, Microsoft Hololens, Google Glass, Valve Index, HTC Vive, Raptor AR, Psious (Amelia), Magic Leap, Epson Moverio, and IKEA Place AR. Next, the specifications for the devices were reviewed and highlighted. Then we performed a security evaluation of the websites for each device and identified security vulnerabilities therein using the website application BuiltWith¹ to take an educated guess on the backing architecture of the websites. BuiltWith provides information on what a website is built with. It enables the tracking of widgets, frameworks, content delivery networks, web standards, web servers, and others. Finally, we performed a comparative security evaluation for each device across their authentication mode, user profiling, access control, privacy, and database security by investigating the information shared by the manufacturer, articles, and blog reviews.

A. Devices and Applications Selection and Specifications

Our search through the web for AR/VR devices and applications used explicitly in gaming, healthcare, engineering, education, accessibility, retail, energy, and entertainment industry yielded 10 in top devices and applications: Oculus Quest2, Microsoft Hololens, Google Glass, Valve Index, HTC Vive, Raptor AR, Psious (Amelia), Magic Leap, Epson Moverio, and IKEA Place AR. We were particularly interested in the IKEA Place AR application due to the significance of the application in the retail industry. The app enables users to view and assess

their apartment furniture virtually without using a head mount display or smart glass. In the subsections below, we highlight the various specifications of these devices, the industry where they are primarily utilized, and how they are used.

1) *Oculus Quest2*: The Oculus Quest2 is developed by Oculus, a company owned by Meta, and is used in many industries, from gaming to fitness to live events and more. The Oculus Quest2 supports 6 degrees of head and hand tracking with an integrated microphone and speakers. Two hand controllers are included with the VR headset and use the Qualcomm Snapdragon XR2 platform with 6GB RAM and a near 4K display.

2) *Microsoft Hololens*: Microsoft Hololens was developed by Microsoft and is used in industries such as policing and medical training. The Hololens2 specifically is “an untethered holographic computer” according to Microsoft and runs on Windows Holographic OS, which is based on Windows 10. In addition, a user account from Microsoft is required to use the Hololens2, and the device can track both hands and eyes and can be controlled via voice commands. The Hololens 2 has 4 cameras to track head movements, 2 infrared cameras to track eye movement, depth sensors, an accelerometer, gyroscope, magnetometer, and a camera that can record images and video. It has 6 degrees of freedom tracking, mixed reality capture, and a real-time environment mapping for users to use. The underlying architecture involves a Qualcomm Snapdragon chip with 4GB of RAM, 64GB of storage, WiFi and Bluetooth connectivity, and a USB-C connection. The lithium-ion battery typically lasts between two and three hours, and the device is fully functional while charging.

3) *Google Glass*: The Google Glass Enterprise Edition is developed by Google and is used in industries such as healthcare and energy as well as technology. The device is marketed as a way for employees to work “smarter, faster and safer” with the ability to use voice commands and be worn all day. It runs on a Qualcomm Snapdragon chip running Android Open Source Project 8.1 (Oreo) with 3GB of memory and 32GB of flash storage. Google Glass is WiFi and Bluetooth enabled and has a Camera with an 83 degree field of view that can record images and video. In addition, the device can capture touch via a gesture touchpad and recognize voice commands via 3 built-in microphones in addition to an accelerometer, gyroscope, and magnetometer.

4) *Valve Index*: The gaming platform Steam develops the Valve Index. It is primarily used as a gaming device meant to be used directly with the Steam gaming platform. Depending upon the bundle bought, the Valve index includes integrated headphones and external controllers to be used as hand and gesture controls. Furthermore, the system requirements include a processor with a dual-core with hyper-threading, 8GB RAM in memory, NVIDIA GeForce GTX 970 or AMD RX48 as the graphics card.

5) *HTC Vive*: The HTC Vive was developed by Vive and is mainly used in the gaming industry and primarily on the Steam platform. It comes with HTC Vive wand controllers, and the tracking area is 3.5m by 3.5m with a 110 degree

field of view. The required computer specifications are that the device must run on an Intel or AMD chip running on Windows 7 SPI or newer with 4GB of memory and both an HDMI or DisplayPort and a USB 2.0. In addition, it has an OLED display with a built-in mic, an accelerometer, and a gyroscope, but no built-in audio.

6) *Raptor AR*: Raptor AR Glasses are developed by Eyesight and are used as augmented reality cycling glasses to give riders an edge while being unobtrusive. Raptor AR Glasses use patented Eyesight BEAM projection technology to enhance the world around a user and are designed for an outdoor environment. The heads-up display can be activated and deactivated at will and is meant to fit seamlessly into the riders' current training. The Raptor runs on a Qualcomm Snapdragon chip running Android 5.1.1 Lollipop with 2GB of memory and comes in either 16 or 32GB of storage. In addition, it comes equipped with a front-facing camera, an internal speaker, 2 microphones, an accelerometer, gyroscope, magnetometer, barometer, and proximity sensor to detect when the Raptor is equipped.

7) *Psious*: Psious was a VR headset that was a leader in the therapy space, but it appears that they recently re-branded as Amelia Virtual Care while gathering data for this paper. The data gathered during our exploration straddles the space between the re-brand, and therefore we will continue to refer to the device as Psious. However, our data comes from several sources, including Wayback Machine, the internet archive. Psious VR headset is specifically designed for use in the psychology and therapy industry and is used in conjunction with the Psious virtual care platform. The VR headset runs on a Qualcomm Snapdragon chip with 4GB of memory and a 3K resolution screen. In addition, it comes with a sensor that is not attached to the headset but is meant as part of the treatment and measures galvanic skin response or electrodermal activity of the fingers, which Mindfield Biosystems Ltd develops.

8) *Magic Leap*: The Magic Leap VR device is used in medical, manufacturing, defense, and the public sector. The device runs on Lumin Os with 8GB of memory and 128GB of storage. It has a 50 degree field of view with 1.3M pixels per eye and a camera that can capture images and video. The device also has built-in stereo speakers but does not come equipped with headphones. Additionally, there is a degree of freedom control that comes with the device that can provide haptic feedback and hand tracking, while the VR headset tracks eye movement and can respond to voice commands.

9) *Epson Moverio*: The Moverio smart glasses are a cutting-edge accessibility tool and augmented experiences in art and manufacturing, automotive, and logistics. Moverio has three different lines of AR glasses. However, the focus will be on the Moverio BT-40S with an intelligent touch controller. The glasses have a 34 degrees field of vision with an OLED micro-projection and transparent binoculars. The device runs on a Qualcomm SXR chip running Android 9 with 4GB of memory and 64GB of storage.

10) *IKEA Place AR*: There are numerous apps now that use AR to augment the retail experience. However, we will focus

on IKEA's retail AR app allowing users to place furniture in their spaces. Users can scan through over 2000 IKEA products in the online IKEA database, and using IOS's ARKit technology; the app can assess the room's dimensions and scale products with 98% accuracy. IKEA Place AR is used mainly in the retail and design industries, where consumers and designers can place furniture or other items in their spaces without having to purchase the item and then try to figure out whether or not it fits in with the rest of the decor.

IV. RESULTS AND DISCUSSIONS

In this section, we provide our findings based on the in-depth evaluation of the devices in web page security analysis, authentication, user profiling, access control, privacy, and database security, and informed their comparative analysis.

A. Web Page Security Analysis

Each primary website of the AR/VR devices and application in question is an extension of the devices' usage. Users often go to these AR/VR devices' websites for login, help, or find out more information, so it is just as important to look at the underlying issues that could be present and consider that when doing a security evaluation. Using BuiltWith, we collected information about the underlying frameworks in the web pages. We queried the National Institute of Standards and Technology (NIST) ² to search for potential CVEs. NIST provides a National Vulnerability Database, uniquely identifying vulnerabilities alongside the CVE. It is noted that these vulnerabilities, when exploited, negatively impact confidentiality, integrity, or availability.

The primary websites that were queried using this Builtwith can be found in Table I. The data from the BuiltWith was then looked at by one of the researchers who is also a software engineer and technologies were picked out and queried in NIST to search for potential CVEs. For example, the Oculus Quest2 website had 2 possible CVEs relating to a payment platform called PayU. The Microsoft Hololens website had a total of 7 possible CVEs, all relating to the open graph protocol. The Google Glass website had a total of 2409 possible CVEs with 1039 relating to AJAX, 24 relating to Angular, 1339 relating to QUIC, and 7 relating to the open graph protocol. The Valve Index website had 107 possible CVEs relating to Nginx. The HTC Vive website had a total of 117 possible CVEs, the Raptor website had a total of 36,044 possible CVEs, while Magic Leap, IKEA, and Epson Moverio's websites had 4, 59, and 2514 respectively.

B. Authentication in AR/VR Technologies

The primary goal of authentication is to verify the authenticity of the users of the system and their accounts [6]. In AR/VR devices and applications, it is highly essential to manage access to the devices and third-party applications and also access to shared spaces in virtual environments [17]. The devices in our research implemented different authentication

²<https://nvd.nist.gov/vuln/search>

approaches, including knowledge-based authentication, token-based authentication, biometric authentication, social authentication, and even zero authentication. 6 of the devices implemented pin/passwords: Google glass, Valve index, Raptor, Psious, Hololens, HTC Vive, and Epson Moverio. Hololens and Epson Moverio have an optional biometric authentication option. For Moverio, the biometric authentication layer utilizes a palm-based biometric authentication, while Hololens biometric authentication is achieved through iris signatures. Magic Leap utilizes token-based authentication, which utilizes a two-step verification process that involves users visiting the Magic Leap Identity portal and entering their email address which generates a token that is sent to the user's email address for the user to complete the authentication process [23]. Raptor and HTC Vive implement Social authentication as an optional layer. At the same time, Oculus Quest's primary and only layer of authentication is social authentication, which only allows Facebook logins.

While single authentication option is used in Google glass, Psious, Magic Leap, Oculus Quest, and Valve Index. Raptor, Hololens, HTC Vive, and Epson Moverio have provided multi-factor authentication for users to log in to the devices. Hololens, in particular, provides up to 6 authentication options which include; Azure web credential provider, Azure Authenticator App, Biometric (Iris), FIDO2 Security key, PIN, and Password.

C. User Profiling in AR/VR Technologies

User profiling involves collecting specific information associated with a user to create a digital representation of the user's identity [14]. The user profile created for a user in a virtual or augmented reality environment determines their interactive participation behavior.

To assess user profiling in AR/VR technologies, we were interested in knowing the basic information required to create a user profile in an AR/VR environment and the steps required to remove a user profile from the AR/VR environment. We segmented the devices into specific groups to highlight the information required to create a user profile; Social Access and Basic User Information. Raptor, Oculus Quest and HTC Vive utilize social access, an option that allows users to create a profile with their social media account. In this case, it is hard to figure out the information shared between the device and the social media platform to initiate the creation of a user profile [1]. In the other devices; Valve index, Hololens, Google glass, Raptor, and Magic Leap, similar basic information is collected, including some or all of the following; user's name, date of birth, telephone number, country of residence, as well, as an email address. The Microsoft Hololens can be accessed by only a Microsoft account and Google glass only by a Google account which eventually shares data from the account into the device [8], [13]. For the Valve index, the user could create an account without revealing personal identifying information such as their real name. However, user profile deletion in these devices or applications is not as straightforward as user creation. For example, only Oculus

provides the option for users to delete their user account or profile from the application. Others require the user to initiate a request through the support team for account deletion.

Hololens and Google glass did not explicitly provide information regarding user account deletion; instead, the term "device reset" is used. This means that the AR/VR device can be reset in a case where the information of an already existing user would be erased from the device. However, this does not necessarily erase the user's account in the application. In table I, we presented the respective representation of the user account deletion in our selected AR/VR technologies.

D. Access Control in AR/VR Technologies

Access control provides a layer of mediation to resources and data in a system and decides what level of access a user is granted [25]. It eventually prevents activities that could lead to a breach of security [24]. In the context of our research, we were interested in understanding multi-user access in AR/VR technologies. Our findings identified devices utilizing a single-user, multi-user, and devices that do not provide such info. For example, Google glass exhibits a single user login whereby an attempt to have another user access the smart glass or existing user to switch accounts would require the factory reset of the device. We could not find any multi-user access information for Raptor and Psious; however, single-user access is confirmed. Multi-user accounts generally require an administrator to configure the permissions for the other added accounts. Hololens, Oculus, Valve index, Magic Leap, Epson Moverio, and HTC Vive support multiple user access. For Hololens, this is restricted to Azure AD accounts. An Hololens account can be linked with as much as 64 accounts, which share resources across or within the apps. The admin grants permissions for certain users to access limited apps on the device. This feature is available to organizations utilizing AD tenants. Oculus Meta Quest 2, on the other hand, allows up to 4 accounts.

E. Privacy in AR/VR Technologies

Privacy is about safeguarding user identity while using a product or service [27]. It is the right they possess over the usage of their information [12]. We wanted to understand the methods of data collection of the AR/VR technologies, what the companies manufacturing them are utilizing, what the data is being used for, how long they retain user data, their cookie usage, data shared with third parties, and most importantly if they have functionality that allows users to opt-out of the information shared with third parties without writing an official request to them.

We found that all the devices collect information such as user's technical system information, account data, transaction and payment data, user's posts, comments or follow in any of the content and services, information sent through chat, voice interactions, information users provide when they request information or support, the information provided when participating in competitions, contests, and tournaments or responding to surveys, device information, game statistics, IP

TABLE I
DETAILS OF AR/VR TECHNOLOGIES EVALUATED ALONG WITH THE CVE COUNT, AVAILABLE AUTHENTICATION SCHEME, ACCESS CONTROL, DATA SECURITY MECHANISM AND THE WEB URL.

Technology	CVE Count	Authentication	Account Deletion	Access Control	Data Security	URL
Oculus Quest	2	Social	App	Multi user	AES-256 XTS	www.oculus.com/quest-2/
Hololens	7	Pin/password, Biometric	Device	Multi user	TPM e.t.c.	www.microsoft.com/en-us/hololens/
Google Glass	2409	Pin/password	Device	Single user	No info	www.google.com/glass/start/
Valve Index	107	Pin/password	Upon request	Multi user	No info	www.store.steampowered.com/valveindex
HTC Vive	117	Social	Upon request	Multi user	No info	www.vive.com/eu/product/vive/
Raptor AR	36044	Pin/password, Social	No info	Single user	No info	www.eversight.com
Psious (Amelia)	39	Pin/password	No info	Single user	No info	www.professional.ameliavirtualcare.com
Magic Leap	4	Token-based	Upon request	Multi user	RSA-3072, SHA-512	www.magicleap.com/
Epson Moverio	59	Pin/password, Biometric	No info	Multi user	No info	www.moverio.epson.com/
IKEA Place AR	2514	No authentication	No user account	No user account	No info	www.ikea.com/

address, crash reports, and system activity. These data are used to update, secure and troubleshoot.

Only Psious and IKEA indicated in their privacy policy how long they keep and process user's data, and it stated that the data would be maintained as long as necessary to meet the purpose of collection. In addition, all devices utilize cookies for authentication, security, advertising and insights, performance, and analytics and research. Although only Psious did not indicate its usage of cookies in its privacy policy, we realized cookie was used on their website. Furthermore, the AR/VR manufacturers have also informed that this cookie information is shared with third parties such as Google Analytics.

Oculus, Hololens, Google Glass, and IKEA allowed the user to adjust the data being utilized, at least to some extent. Magic Leap does not currently respond to "do not track" signals or other mechanisms that might enable users to opt-out of tracking on their services. With Oculus, users can turn off sharing usage statistics and crash reports. The opt-out page on Microsoft enables users to control their data use. Users can decide to receive promotional emails, SMS messages, telephone calls, and postal mail from Microsoft, and access and clear some of the data through the Microsoft privacy dashboard. Users can customize the data being utilized through Google's Activity controls, Ad settings, and Google dashboard. User profiles in IKEA also have options to control the data usage.

F. Database Security in AR/VR Technologies

Data security includes protecting digital information or data from unauthorized access throughout its lifecycle [9]. It may include encryption, hashing, and tokenization practices to ensure data protection in a system [4]. Database security involves implementing measures to secure database manage-

ment systems from illegitimate use and malicious cyber threats and attacks [5]. In the context of our research, we paid close attention to AR/VR technologies that shared information about some particular strategy that has been put in place to secure users' data in their system. Only Oculus, Hololens, and Magic Leap provided information about how data security is attained. With Oculus, data is encrypted at rest on the headsets using AES-256 XTS, and, if desired, administrators can lock each headset with a PIN.

In addition, TLS 1.2 and TLS 1.3 protocols are used to encrypt data transmitted between the headset and backend servers. Magic Leap mentioned that all app installations are designed to be signed using RSA-3072 and SHA-512. Also, the user data saved in the app is encrypted with the user's private key, so when the device is locked, the user's private data is presumed inaccessible to any attacker. Microsoft Hololens shared quite an extensive list of the security protocols in place. This includes; the use of BitLocker to encrypt user drive, Trusted Platform Module (TPM) to enhance security and privacy through encryption and decryption, Microsoft Defender SmartScreen to help in malware and phishing detection, Digital Signing code to protect the device from untrusted apps, malware and viruses, and Unified Extensible Firmware Interface (UEFI) Secure Boot to prevent attacks through Bluetooth, WiFi, and Ethernet. Also, it is noteworthy to mention a Steam VR breach discovered by Alex Radocea and Philip Pettersson in 2019, which was unveiled at the Montreal Recon conference ³. The researchers were able to project images into a victim's headset depicting the possibility of altering their virtual world, and this could make them create a worm that could spread by infecting anyone in a chat room.

³<https://thevrheadset.com/vrchat-and-steam-vr-allow-hackers-to-control-your-pc/>

V. CONCLUSION

AR/VR has demonstrated the capability to integrate into several industries and provides an immersive experience to the users. Our analysis of these technologies, i.e., both head mount displays, smart glasses and applications, showed that the web pages for these devices have CVEs that could grant attackers access. We also found out that most of these devices or applications do not provide multi factor authentication for the users, and there are mostly no direct means for users to remove their profile on these devices or applications. Furthermore, we realized that most privacy policies are flawed as there is no clarity regarding the data shared with third parties. Only a few of these devices provide a means for users to customize the data being accessed. We provided recommendations on the way forward to provide a secured and enhanced experience for the user when utilizing these devices.

VI. FUTURE WORK AND LIMITATIONS

This study reported in this paper examines a small subsection ($N = 10$) of the overall VR and AR market. We provide a generic overview from the privacy and security front by analyzing the commonly available technologies. However, we plan to add other devices and applications for our detailed analysis as an extension of this work to provide an overview from a quantitative perspective and explore the types of interactions that a user could have. In the future, we would like to look into a more comprehensive array of devices and applications, and perform a more in-depth security and privacy analysis using multiple approach that involves pen testing or static and dynamic code analysis.

VII. ACKNOWLEDGEMENT

We want to acknowledge the Inclusive Security and Privacy-focused Innovative Research in Information Technology (In-Spirit) Lab at the University of Denver. Additionally, we want to thank Evan Dybvig and Madi Fawcett for their input in the research, data collection, and analysis. Any opinions, findings, conclusions, or recommendations expressed in this material are solely those of the authors.

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